Evaluating Fast Matrix Multiplication Algorithms In Rust

Computational Laboratory Project Presentation

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Introduction

- **Matrix Multiplication:** One of the most fundamental computations in numerical linear algebra and scientific computing.
- **Fast Algorithms:** Perform asymptotically fewer floating-point operations and require less data communication than the classical $O(N^3)$ algorithm.
- **Project Goals:**
- Implement fast matrix multiplication using arbitrary CP tensor decompositions in Rust.
- Evaluate native Rust linear algebra library (`faer`) against optimized vendor GEMMs (Intel `MKL dgemm` via FFI).
- Handle general dimensions via memory-efficient **dynamic peeling** instead of zero-padding.
- Benchmark sequential and parallel execution using multiple scheduling strategies.

Tensors & Outer Products

Definition: 3-Way Tensor

A 3-dimensional (or 3-way) tensor is a multidimensional array $T \in \mathbb{R}^{M \times N \times P}$. Its slices obtained by fixing one index are matrices. For example, the k-th frontal slice of T is the matrix X_k.

Definition: Outer Product

For vectors $u \in \mathbb{R}^M, v \in \mathbb{R}^N, w \in \mathbb{R}^P$, the rank-1 outer product tensor $T = u \circ v \circ w \in \mathbb{R}^{M \times N \times P}$ has entries:

$$x_{ijk} = u_i v_j w_k$$

The rank of a tensor T is the minimum number of rank-1 tensors that generate T as their sum.

Low-Rank Tensor Decompositions

Definition: CP Decomposition

The Canonical Polyadic (CP) decomposition of rank R of a tensor T is:

$$T = [\![U, V, W]\!] = \sum_{r=1}^{R} u_r \circ v_r \circ w_r$$

where $U \in \mathbb{R}^{M \times R}$, $V \in \mathbb{R}^{N \times R}$, and $W \in \mathbb{R}^{P \times R}$ are factor matrices with columns u_r, v_r, and w_r.

Definition: Mode-k Product

The mode-k product between a tensor $T \in \mathbb{R}^{N_1 \times N_2 \times N_3}$ and a matrix $U \in \mathbb{R}^{R \times N_k}$ for $k \in \{1, 2, 3\}$ is denoted by $Y = T \times_k U$. Its matricized representation is:

$$Y_{(k)} = U T_{(k)}$$

Bilinear Forms via Low-Rank Decompositions

- A bilinear form (between finite dimensional vector spaces X, Y and Z) is a mapping $f : X \times Y \to Z$ which can be represented by a tensor T as

$$f = T \times_1 x^T \times_2 y^T$$

- Using a CP decomposition $T = [\![U,V,W]\!]$ of rank R:

$$f(x, y) = \sum_{l=1}^{R} (u_l^T x) \cdot (v_l^T y) w_l$$

- Evaluating $f(x, y)$ requires exactly R active scalar multiplications and $O(R)$ additions.
- For matrix multiplication $C = AB$, representing the form $\langle M,N,P \rangle$, we have:

$$\operatorname{vec}(C^T) = T \times_1 \operatorname{vec}(A)^T \times_2 \operatorname{vec}(B)^T$$

- Given a CP decomposition of $T = [\![U,V,W]\!]$, we get:

$$\operatorname{vec}(C^T) = \sum_{l=1}^{R} (s_l \cdot t_l) w_l$$

where $s_l = u_l^T \operatorname{vec}(A)$ and $t_l = v_l^T \operatorname{vec}(B)$.

Strassen's Algorithm & CP Decomposition

- **Strassen's Algorithm** for $\langle 2,2,2 \rangle$ uses $R = 7$ multiplications and 18 additions:

$$U = \begin{pmatrix} 1 & 0 & 1 & 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 & 0 & 0 & -1 \end{pmatrix},$$

$$V = \begin{pmatrix} 1 & 1 & 0 & -1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 1 \\ 1 & 0 & -1 & 0 & 1 & 0 & 1 \end{pmatrix},$$

$$W = \begin{pmatrix} 1 & 0 & 0 & 1 & -1 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 & 0 \\ 1 & -1 & 1 & 1 & 0 & 0 & 0 \end{pmatrix}$$

- This reduces arithmetic complexity from $O(N^3)$ to $O\left(N^{\log_2 7}\right) = O\left(N^{2.81}\right)$.
- Fast algorithms can be constructed for general base cases $\langle m,n,p \rangle$ using CP decompositions of rank R.

Summary of Fast Matrix Algorithms

Algorithm Base Case	Multiplications (R)	Classical (mnp)	Speedup per Step
$\langle 2,2,2 \rangle$ (Strassen)	7	8	14%
$\langle 2,2,3 \rangle$	11	12	9%
$\langle 2,2,4 \rangle$	14	16	14%
$\langle 2,2,5 \rangle$	18	20	11%
$\langle 2,3,3 \rangle$	15	18	20%
$\langle 2,3,4 \rangle$	20	24	20%
$\langle 2,4,4 \rangle$	26	32	23%
$\langle 3,3,3 \rangle$	23	26	17%
$\langle 3,3,4 \rangle$	29	36	24%
$\langle 3,3,6 \rangle$	40	54	35%

Dynamic Peeling

- Traditional fast algorithms require dimensions to match the base case power.
- **Dynamic Peeling:** A memory-efficient alternative to zero-padding.
- Split A and B into core components and peeled boundary vectors/scalars:

$$A = \begin{pmatrix} A_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}, B = \begin{pmatrix} B_{11} & b_{12} \\ b_{21} & b_{22} \end{pmatrix}$$

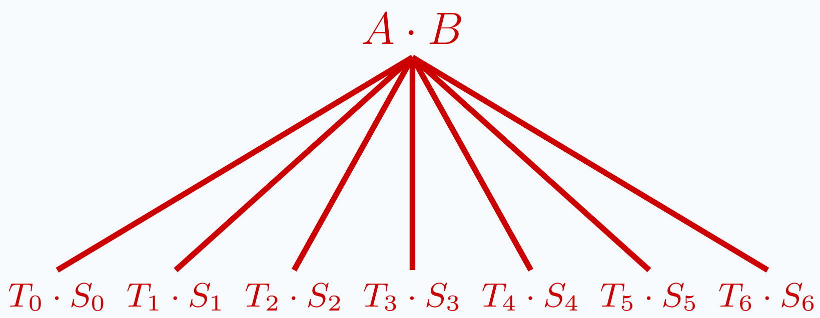
- Recombine using a combination of the core product and boundary corrections:

$$C = \begin{pmatrix} C_{11} & c_{12} \\ c_{21} & c_{22} \end{pmatrix} = \begin{pmatrix} A_{11}B_{11} + a_{12}b_{21} & A_{11}b_{12} + a_{12}b_{22} \\ a_{21}B_{11} + a_{22}b_{21} & a_{21}b_{12} + a_{22}b_{22} \end{pmatrix}$$

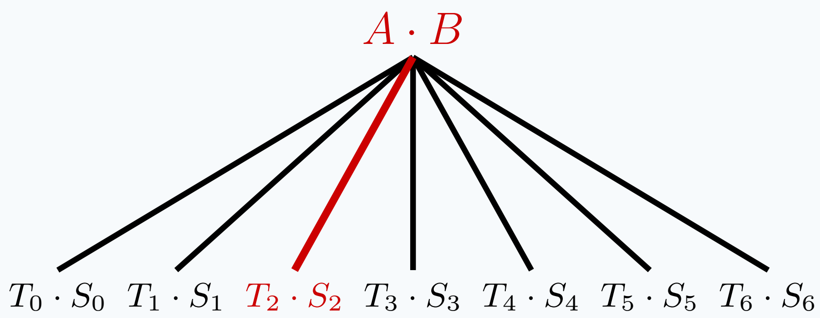
- Leaf-node $A_{11}B_{11}$ is solved recursively, while all boundary corrections are consolidated into single, large GEMM calls to maximize cache locality and vendor BLAS efficiency.

Shared Memory Parallel Scheduling

Breadth-First-Search (BFS)



Depth-First-Search (DFS)



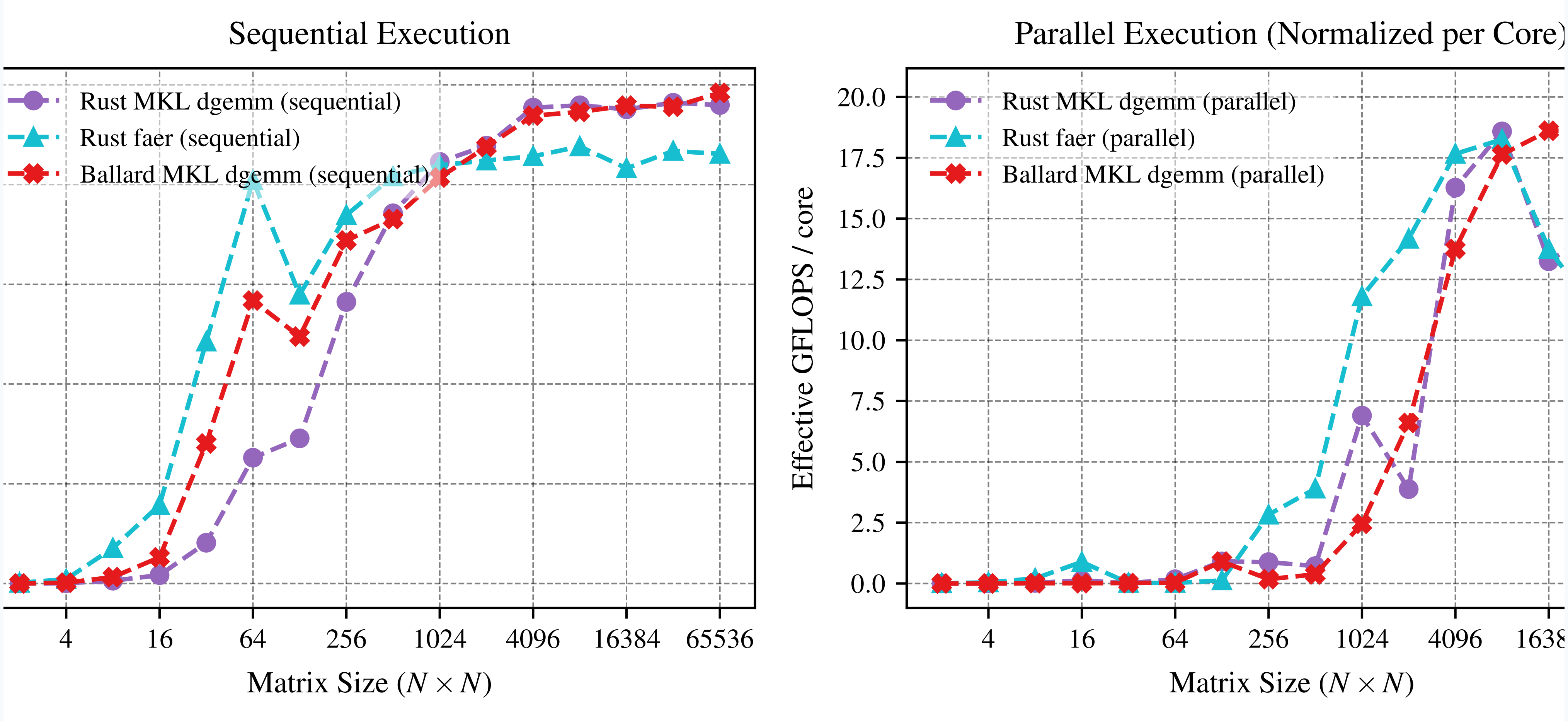
Execution Platform & Setup

- **Hardware:** One node of a dual-socket AMD EPYC 7763 server (64 physical cores, 2 TB RAM).
- **Software environment:**
- GCC 13.2.0, Intel OneAPI compilers 2025.0.4, Intel OneAPI MKL.
- Rust Nightly (2026-06-23) utilizing LLVM 22 for optimized AVX2 and FMA generation.
- **Linear Algebra Backends:**
- **Intel MKL dgemm:** Vendor library called via custom FFI wrapper.
- **faer matmul:** Native Rust high-performance linear algebra library.
- **Metric:**

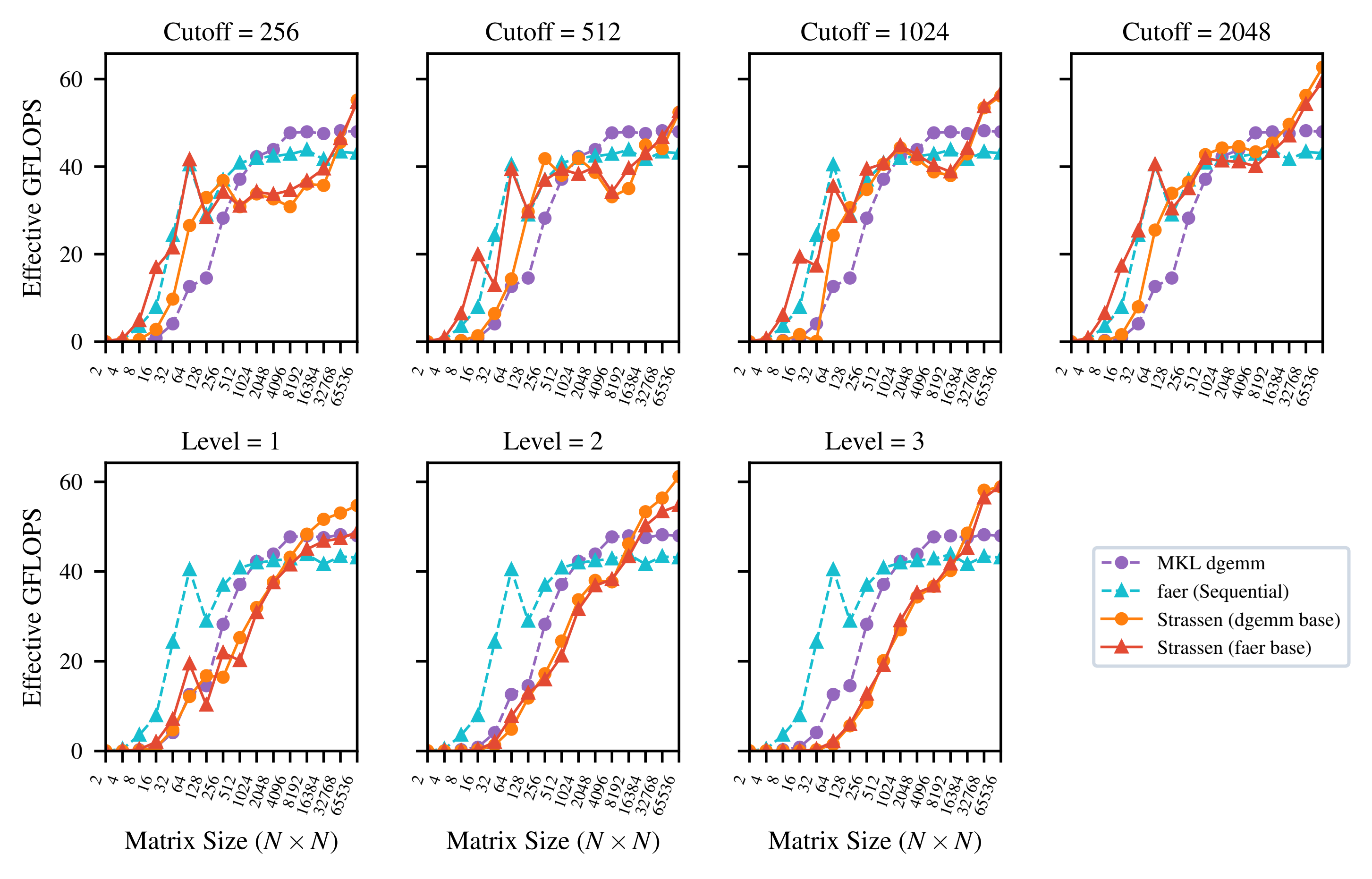
$$\text{Effective GFLOPS} = \frac{2MNP - MP}{\text{time in seconds} \cdot 10^9}$$

which normalizes performance based on the classical floating-point operation count.

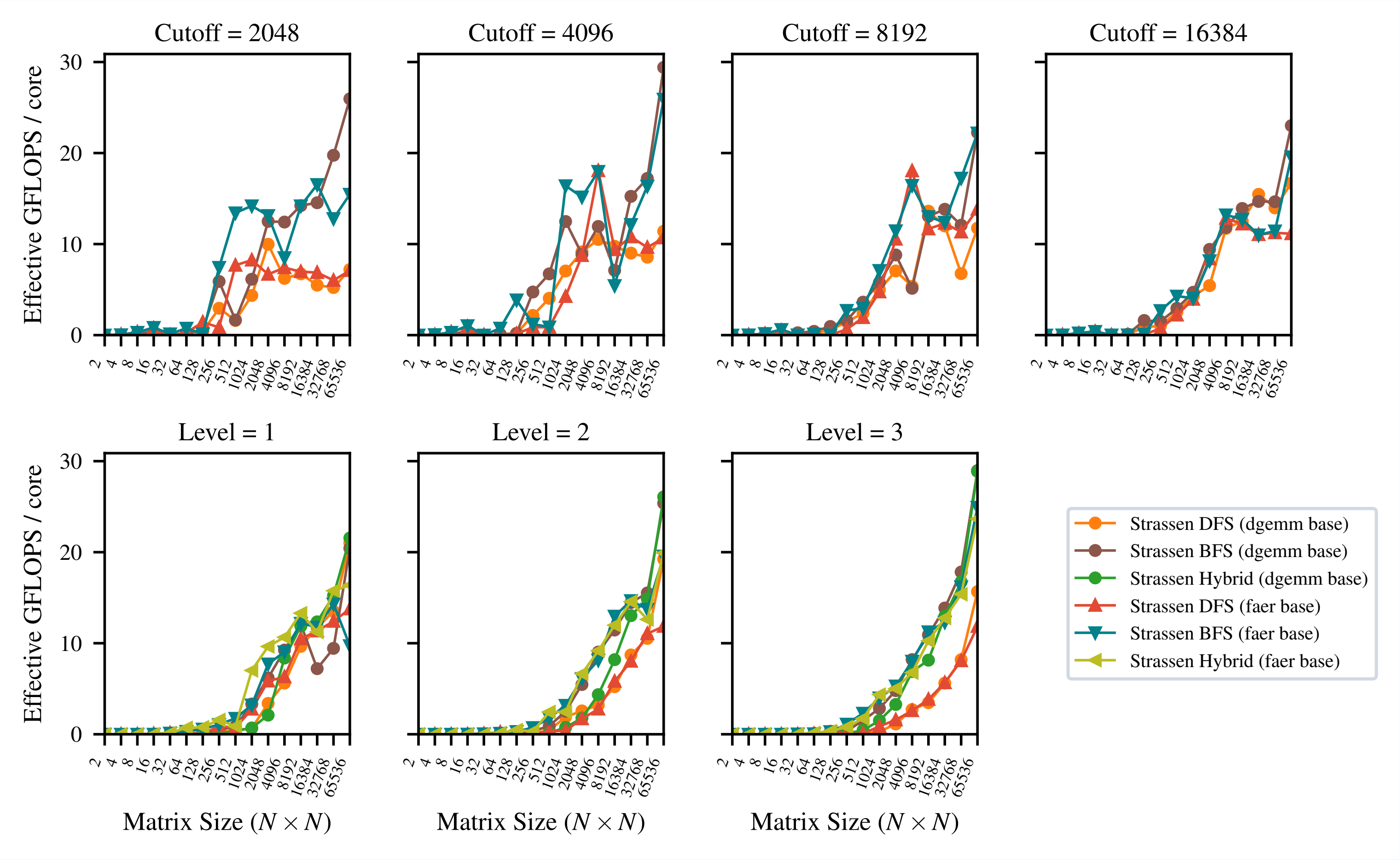
Performance of Base GEMM Libraries



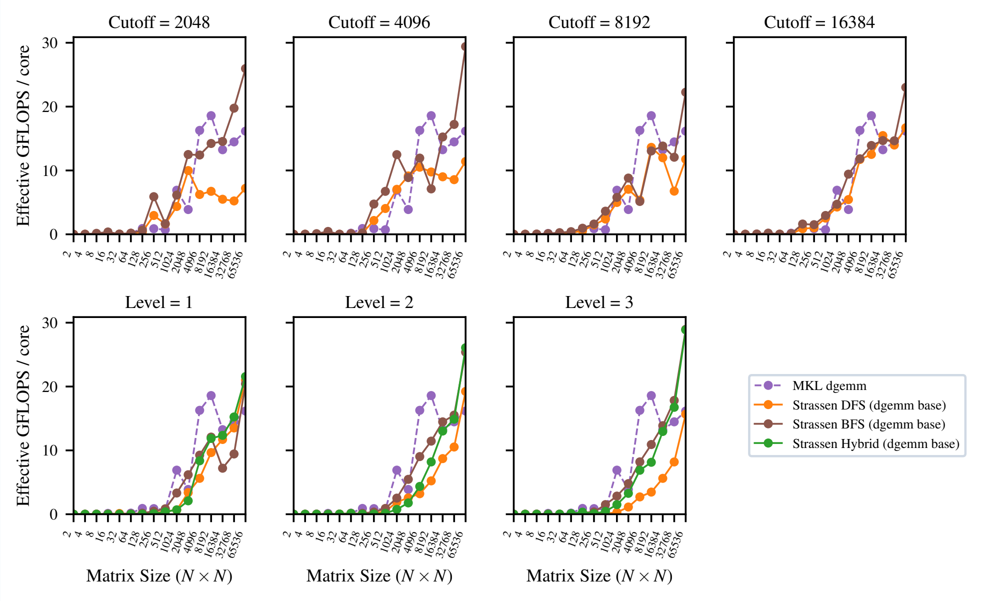
Sequential Crossover Point

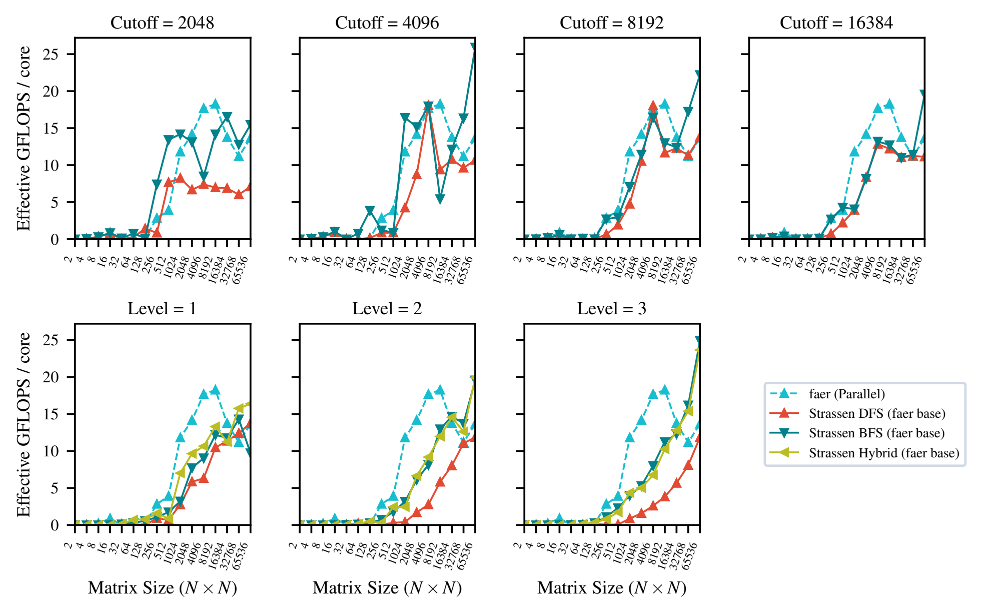


Parallel Crossover (Normalized per Core)

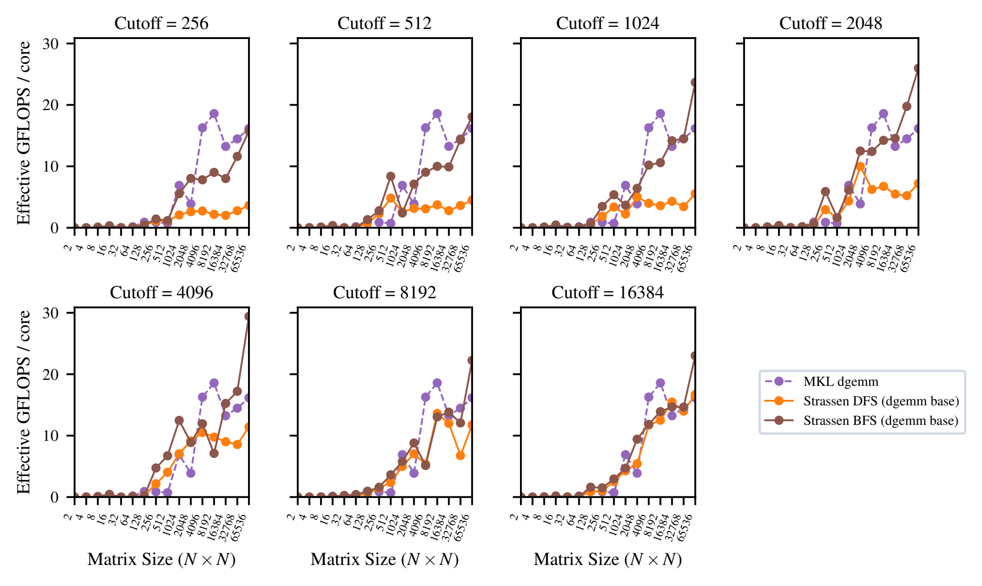


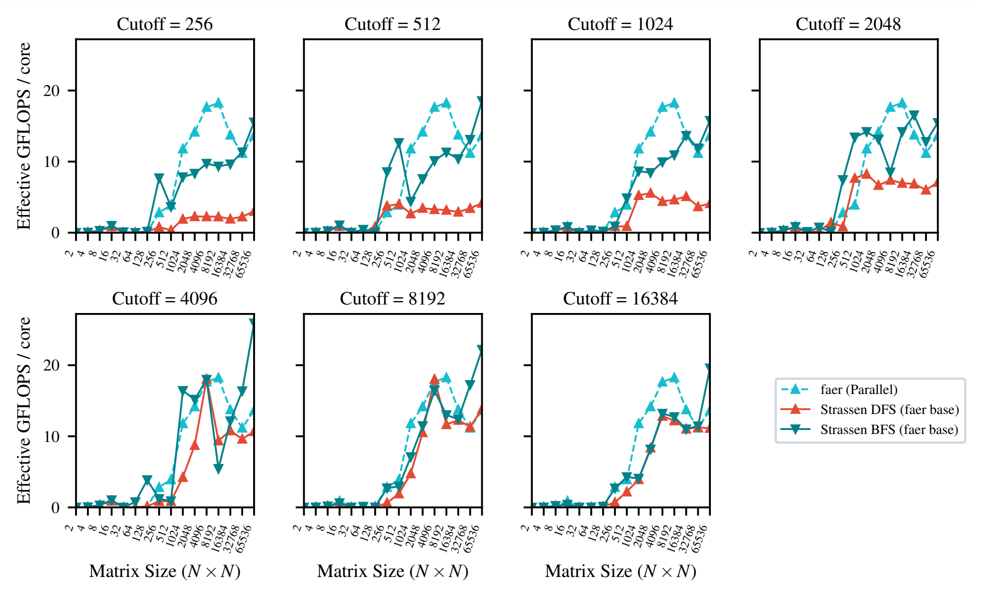
Parallel Cutoffs: MKL vs. Faer Bases



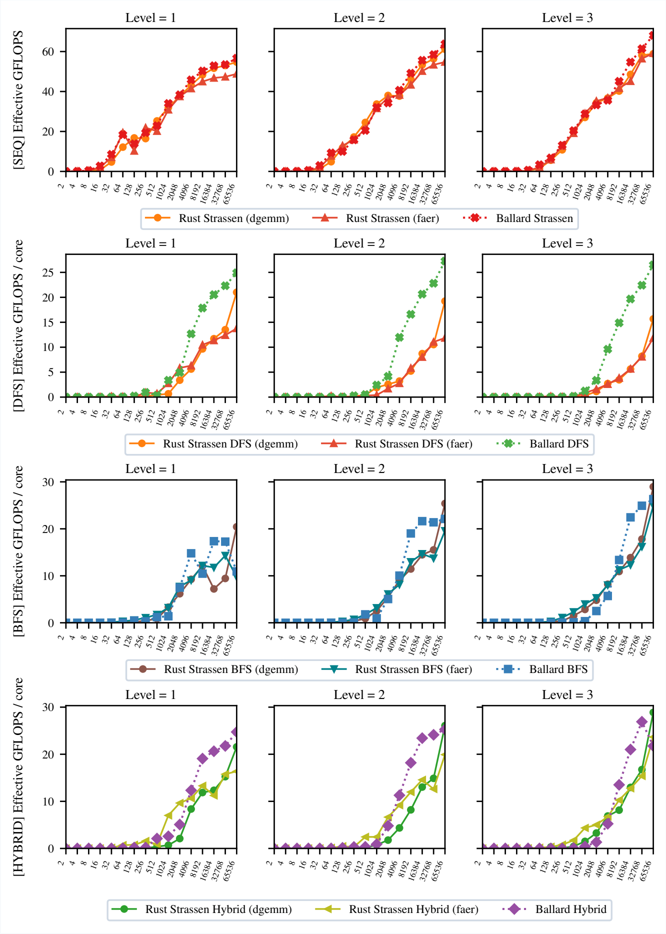


Finding the Optimal Size Cutoff





Comparison with Benson & Ballard C++ Reference



Conclusions & Future Work

Key Takeaways:

- Fast matrix multiplication algorithms successfully outperform optimized vendor/native GEMM libraries in both sequential and parallel execution.
- Crossover points are highly practical ($N = 8192$ sequential, $N = 16384$ parallel).
- Rayon's work-stealing scheduler provides superior load-balancing and scaling compared to OpenMP at deeper levels of recursion.

- **Future Directions:**
- **Profiling & SIMD:** Finer refactoring of matrix split/addition routines using SIMD vectorization.
- **Other GEMM Backends:** Add support for OpenBLAS, BLIS, or other open-source libraries.
- **Distributed/GPU version:** Port to GPU (CUDA/WebGPU) or distributed clusters to scale beyond $N = 65536$.
- **Architectural Comparisons:** Evaluate on high-core-count Intel Xeon processors.